

Aswathi M

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Summary

Enthusiastic Computer Science and Engineering student specializing in cybersecurity with experience in ML-based anomaly detection and web development. Skilled in Python and Java, with knowledge of C, C++, data structures, algorithms, and operating systems. A motivated learner with an interest in cybersecurity, actively involved in technical and social activities, and eager to gain practical experience through real-world applications.

Education

Amrita Vishwa Vidyapeetham (2023 – Present) – Kerala

B.Tech, Computer Science and Engineering with Cybersecurity - 78.8%

Sreekrishnapuram Higher Secondary School (2022) – Kerala

Biology Science - 96.5%

Sreekrishnapuram Higher Secondary School (2020) – Kerala

SSLC - 99.0%

Projects

- **Federated Learning–based Secure Access Control in IoMT:** Architected a privacy-preserving federated learning framework for IoMT, enabling secure API access and real-time anomaly detection using Zero Trust and RBAC.
- **Proactive Plant Disease Prediction System:** Developed a dual-modality ML/DL system using XGBoost and MobileNetV2, achieving 98.9% accuracy and enabling proactive disease prediction using soil and image data.
- **HomeScope – Interior Design Website:** Created a responsive interior design website with interactive UI/UX features and purchase options for customizable designs.

Technical Skills

Programming: Python, Java, C, C++

Web & Backend: HTML, CSS, Flask, REST APIs, JWT

Machine Learning: Scikit-learn, Pandas, NumPy

Security: Zero Trust Architecture, RBAC, API Security

Database & Tools: PostgreSQL, Git, Linux

Achievements

- **Federated Learning–based Secure Access Control in IoMT:** Presented at the 11th IEEE International Symposium on Smart Electronic Systems (iSES) 2025.
- **Proactive Plant Disease Prediction System:** Presented at International Conference on Future Tech: AI-Driven Innovations for Sustainable Development Goals (Future Tech 2025).
- Displayed the project at Expo and Poster Presentation – **IIC Amritapuri**.
- Organized and led an educational session on Indian heritage at a neighboring school as part of the **SSR project**.

Certificates

- Attended a workshop on Research Opportunities in Computer Science.
- Cyber Security Foundation Certification, Infosys Springboard – August 2025.

Extracurricular / Volunteering

Creative Skills: Dance, Drawing

Volunteering: Refreshment Coordinator & Committee Member – Amritaprabah Club, Coordinator – MATLAB Workshop, Event Decor Team Member – Vidyut Fest, Volunteer – Amritavarsham Celebration, SSR Day

Spoken Languages

English, Hindi, Malayalam, Tamil